

```
In [3]: import numpy as np
import pandas as pd
```

```
In [4]: human_interaction=pd.read_csv('personality_data.csv')
human_interaction
```

```
Out[4]:
```

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drain
0	4.0	No	4.0	6.0	No
1	9.0	Yes	0.0	0.0	Yes
2	9.0	Yes	1.0	2.0	Yes
3	0.0	No	6.0	7.0	No
4	3.0	No	9.0	4.0	No
...	...	...	...	...	...
2895	3.0	No	7.0	6.0	No
2896	3.0	No	8.0	3.0	No
2897	4.0	Yes	1.0	1.0	Yes
2898	11.0	Yes	1.0	3.0	Yes
2899	3.0	No	6.0	6.0	No

2900 rows x 8 columns

Dataset Information

The title of this dataset is Human Interaction, because we're exploring, different factors, such as, alone time, stage fear, how frequent they attend social events, how frequent they go outside, if they feel drained after socializing, the amount of friends in their circle, and how frequent they post on social media, that contribute to people's personality (extrovert vs introvert).

Human Interaction

The primary purpose of this data is to show how a person's personality influences their daily habits. By looking at these patterns, we can understand why some people need more social time while others need more time alone to recharge their energy.

Basic data exploration

```
In [5]: print(f'The first 5 rows of the dataset:')
human_interaction.head(5)
```

```
Out[5]: The first 5 rows of the dataset:
```

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drained_a
0	4.0	No	4.0	6.0	No

	Stage_fear	Drained_after_socializing	Personality
1	Yes	Yes	Introvert
2	Yes	Yes	Introvert
3	No	No	Extrovert
4	No	No	Extrovert
...	...	...	...
2895	No	No	Extrovert
2896	No	No	Extrovert
2897	Yes	Yes	Introvert
2898	Yes	Yes	Introvert
2899	No	No	Extrovert

2900 rows × 3 columns

In [10]: `human_interaction.select_dtypes(include=['number'])`

Out[10]:

	Time_spent_Alone	Social_event_attendance	Going_outside	Friends_circle_si
0	4.0	4.0	6.0	13.0
1	9.0	0.0	0.0	0.0
2	9.0	1.0	2.0	5.0
3	0.0	6.0	7.0	14.0
4	3.0	9.0	4.0	8.0
...	...	...	...	...
2895	3.0	7.0	6.0	6.0
2896	3.0	8.0	3.0	14.0
2897	4.0	1.0	1.0	4.0
2898	11.0	1.0	3.0	2.0
2899	3.0	6.0	6.0	6.0

2900 rows × 5 columns

The Stage_fear and Drained_after_socializing columns should be converted from categorical to numerical in order to check if there is a correlation analysis between all the data that explains the person personality type

Descriptive Summary

In [11]:

```

hi_summary=human_interaction.describe()
print(hi_summary)
hi_mean=human_interaction['Time_spent_Alone'].mean()
print(hi_mean)
hi_median=human_interaction['Time_spent_Alone'].median()
print(hi_median)
hi_range=human_interaction['Time_spent_Alone'].max()+human_interaction['Time_spent_Alone'].min()
print(hi_range)
hi_std=human_interaction['Time_spent_Alone'].std()
print(hi_std)

```

```
Out[11]:
```

	Time_spent_Alone	Social_event_attendance	Going_outside	\
count	2900.000000	2900.000000	2900.000000	
mean	4.505816	3.963354	3.000000	
std	3.441180	2.872608	2.221597	
min	0.000000	0.000000	0.000000	
25%	2.000000	2.000000	1.000000	
50%	4.000000	3.963354	3.000000	
75%	7.000000	6.000000	5.000000	
max	11.000000	10.000000	7.000000	

	Friends_circle_size	Post_frequency
count	2900.000000	2900.000000
mean	6.268863	3.564727
std	4.232340	2.893587
min	0.000000	0.000000
25%	3.000000	1.000000
50%	5.000000	3.000000
75%	10.000000	6.000000
max	15.000000	10.000000

4.50581600281988
4.0
11.0
3.4411804188954207

The mean size of a friend circle is approximately 6.27, but the standard deviation of 4.23 tells us there is a high degree of variability in the data. While the "typical" person has about 6 friends, some individuals report having 0 friends (minimum), while others have up to 15 friends (maximum). This wide range suggests that the dataset captures diverse social behaviors, likely categorized by the personality types (Introverts vs. Extroverts). For most columns like Social_event_attendance and Going_outside the mean and median are very close. This shows that the data for these behaviors is relatively symmetric and not heavily skewed by extreme outliers. There are no negative values or missing values in the numerical summaries.

Missing or Duplicate Data

```
In [12]: missing_value=human_interaction.isnull().sum()
print(f'There are {missing_value} duplicated values in each column in
human interaction')
```

```
Out[12]: There are Time_spent_Alone          0
Stage_fear          0
Social_event_attendance  0
Going_outside        0
Drained_after_socializing  0
Friends_circle_size    0
Post_frequency        0
Personality           0
dtype: int64 duplicated values in each column in human interaction
```

```
In [13]: duplicate=human_interaction.duplicated().sum()
print(f'There are {duplicate} duplicated rows in human interaction')
```

```
Out[13]: There are 402 duplicated rows in human interaction
```

Data Wrangling

Adding a new column based on another

In [14]: `hi_new=human_interaction.copy()`
`hi_new`

Out [14]:

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drain
0	4.0	No	4.0	6.0	No
1	9.0	Yes	0.0	0.0	Yes
2	9.0	Yes	1.0	2.0	Yes
3	0.0	No	6.0	7.0	No
4	3.0	No	9.0	4.0	No
...	...	...	...	...	...
2895	3.0	No	7.0	6.0	No
2896	3.0	No	8.0	3.0	No
2897	4.0	Yes	1.0	1.0	Yes
2898	11.0	Yes	1.0	3.0	Yes
2899	3.0	No	6.0	6.0	No

2900 rows x 8 columns

In [15]:

```
def drained(Time_spent_Alone):
    if pd.isnull(Time_spent_Alone):
        return 'N/A'
    elif Time_spent_Alone < 5:
        return 1
    else:
        return 2

def fright(Friends_circle_size):
    if pd.isnull(Friends_circle_size):
        return 'N/A'
    elif Friends_circle_size <= 5:
        return 2
    else:
        return 1
```

In [16]: `hi_new['Drained_after_events']=hi_new['Time_spent_Alone'].apply(drained)`
`hi_new['Stage_fright']=hi_new['Friends_circle_size'].apply(fright)`
`hi_new`

Out [16]:

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drain
0	4.0	No	4.0	6.0	No
1	9.0	Yes	0.0	0.0	Yes
2	9.0	Yes	1.0	2.0	Yes
3	0.0	No	6.0	7.0	No
4	3.0	No	9.0	4.0	No
...	...	...	...	...	...
2895	3.0	No	7.0	6.0	No
2896	3.0	No	8.0	3.0	No
2897	4.0	Yes	1.0	1.0	Yes
2898	11.0	Yes	1.0	3.0	Yes
2899	3.0	No	6.0	6.0	No

2900 rows x 10 columns

In [17]: `hi_new.head(11)`

Out [17]:

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drained
0	4.0	No	4.0	6.0	No
1	9.0	Yes	0.0	0.0	Yes
2	9.0	Yes	1.0	2.0	Yes
3	0.0	No	6.0	7.0	No
4	3.0	No	9.0	4.0	No
5	1.0	No	7.0	5.0	No
6	4.0	No	9.0	3.0	No
7	2.0	No	8.0	4.0	No
8	10.0	Yes	1.0	3.0	Yes
9	0.0	No	8.0	6.0	No
10	3.0	No	9.0	6.0	No

The new column Drained after events is based on the person's rating of how much time they spend alone. If they spend less than 5 hours alone, they're considered extroverts who rarely feel drained after socializing. So those who are considered extroverts are given the number 1, in case a correlation table is needed later on. And if they have 5 or less friends they are considered introverts, so they are given the number 2.

```
hi_new.to_csv('Personality_Dataset_filtered1.csv')
```

Filtering the Data

In [19]: `hi_filtered1=hi_new.copy()
hi_filtered1`

Out [19]:

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drain
0	4.0	No	4.0	6.0	No
1	9.0	Yes	0.0	0.0	Yes
2	9.0	Yes	1.0	2.0	Yes
3	0.0	No	6.0	7.0	No
4	3.0	No	9.0	4.0	No
...	...	...	...	...	...
2895	3.0	No	7.0	6.0	No
2896	3.0	No	8.0	3.0	No
2897	4.0	Yes	1.0	1.0	Yes
2898	11.0	Yes	1.0	3.0	Yes
2899	3.0	No	6.0	6.0	No

2900 rows × 10 columns

In [20]: `hi_filtered1.drop(columns=['Drained_after_socializing'], inplace=True)
print('\n\n Data after dropping Drained after socializing column:\n\n')
hi_filtered1`

Out [20]:

Data after dropping Drained after socializing column:

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Friends_circle_size
0	4.0	No	4.0	6.0	13.0
1	9.0	Yes	0.0	0.0	0.0
2	9.0	Yes	1.0	2.0	5.0
3	0.0	No	6.0	7.0	14.0
4	3.0	No	9.0	4.0	8.0
...	...	...	...	...	...
2895	3.0	No	7.0	6.0	6.0
2896	3.0	No	8.0	3.0	14.0
2897	4.0	Yes	1.0	1.0	4.0
2898	11.0	Yes	1.0	3.0	2.0
2899	3.0	No	6.0	6.0	6.0

2900 rows x 9 columns

In [21]: `hi_filtered=hi_filtered1.copy()
hi_filtered`

Out[21]:

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Friends_circle_size
0	4.0	No	4.0	6.0	13.0
1	9.0	Yes	0.0	0.0	0.0
2	9.0	Yes	1.0	2.0	5.0
3	0.0	No	6.0	7.0	14.0
4	3.0	No	9.0	4.0	8.0
...	...	...	...	...	...
2895	3.0	No	7.0	6.0	6.0
2896	3.0	No	8.0	3.0	14.0
2897	4.0	Yes	1.0	1.0	4.0
2898	11.0	Yes	1.0	3.0	2.0
2899	3.0	No	6.0	6.0	6.0

2900 rows x 9 columns

In [22]: `hi_filtered.drop(columns=['Stage_fear'], inplace=True)
print('\n\n Data after dropping Stage fear column:\n\n')
hi_filtered`

Out[22]:

Data after dropping Stage fear column:

	Time_spent_Alone	Social_event_attendance	Going_outside	Friends_circle_size
0	4.0	4.0	6.0	13.0
1	9.0	0.0	0.0	0.0
2	9.0	1.0	2.0	5.0
3	0.0	6.0	7.0	14.0
4	3.0	9.0	4.0	8.0

	Time_spent_Alone	Social_event_attendance	Going_outside	Friends_circle_si
...	...	...	...	...
2895	3.0	7.0	6.0	6.0
2896	3.0	8.0	3.0	14.0
2897	4.0	1.0	1.0	4.0
2898	11.0	1.0	3.0	2.0
2899	3.0	6.0	6.0	6.0

2900 rows × 8 columns

```
In [23]: hi_filtered.to_csv('Personality_Dataset_filtered.csv', index=False)
```

I filtered the data to make the stage fear column and drained after events column numerical and dropped the categorical columns.

Unique Values and Categories

```
In [24]: unique_values=hi_new['Stage_fear'].unique()
print(f'Are there any unique values in the Stage fear column?
{unique_values}')
```

Out[24]: Are there any unique values in the Stage fear column? ['No' 'Yes']

```
In [25]: unique_values1=hi_new['Drained_after_socializing'].unique()
print(f'Are there any unique values in the Drained after socializing
column? {unique_values1}')
```

Out[25]: Are there any unique values in the Drained after socializing column? ['No' 'Yes']

```
In [26]: unique_values2=hi_new['Personality'].unique()
print(f'Are there any unique values in the Personalty column?
{unique_values2}')
```

Out[26]: Are there any unique values in the Personalty column? ['Extrovert' 'Introvert']

Personality column is the best variable for grouping to see the big-picture differences, while the Drained_after_socializing and Stage_fear columns are better for correlation analysis to see how specific anxieties or energy levels impact a person's lifestyle

Gouping and Aggregation

```
In [27]: personality_by_stage_flight=hi_filtered.groupby('Personality').agg({
    'Time_spent_Alone': 'mean',
    'Friends_circle_size': 'mean',
    'Social_event_attendance': 'mean'
})
personality_by_stage_flight
```

Out [27]:

	Time_spent_Alone	Friends_circle_size	Social_event_attendance
Personality			
Extrovert	2.122869	9.095744	5.977850
Introvert	7.027444	3.277465	1.831621

For this data, I learned that there is a distinction in lifestyle hbit between the two personality types. Intoverts spend signiicantly more time alone an have similar friend circles, while Extroverts attend nearly three times as many social events and maintain larger social networks on average.

Summary and Conclusion

One interesting aspect of this dataset is the almost textook contrast between how introverts and extroverts manage their social energy and time. A clear pattern appeared during the initial exploration: introverts report spending over three times as much time alone as extroverts, while extroverts maintain significantly larger friend circles and attend more social events. However, a few questions arise regarding the data's distribution, particularly, the Friends_circle_size column shows some people with zero friends maintain a high Post_frequency on social media, which suggests that digial social engagement doesn't always mirror physical social circles. Additionally, the lack of 'neutral' or 'Ambivert' categories rises the question of whether the data was intentionally simplfied for classification puoses or if individuals were forced to choose between those two options.

Data Visualization

Out [28]:

hi_new

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drain
0	4.0	No	4.0	6.0	No
1	9.0	Yes	0.0	0.0	Yes
2	9.0	Yes	1.0	2.0	Yes
3	0.0	No	6.0	7.0	No
4	3.0	No	9.0	4.0	No
...	...	...	...	...	...
2895	3.0	No	7.0	6.0	No
2896	3.0	No	8.0	3.0	No
2897	4.0	Yes	1.0	1.0	Yes
2898	11.0	Yes	1.0	3.0	Yes
2899	3.0	No	6.0	6.0	No

2900 rows x 10 columns

```
#importing the libraries
import numpy as np
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
plt.figure(figsize=(8,6))
corr_matrix=hi_new.corr(numeric_only=True)
ax=sns.heatmap(corr_matrix, annot=True, fmt='.3f', cmap='GnBu', center=0,
linewidths=0.5, cbar_kws={'label':'Correlatation'})
plt.title('Correlation Heat Map of Numeric Features')
plt.savefig('Project2_Graphs/Heatmap.jpg')
plt.show()
```

Out [30]:



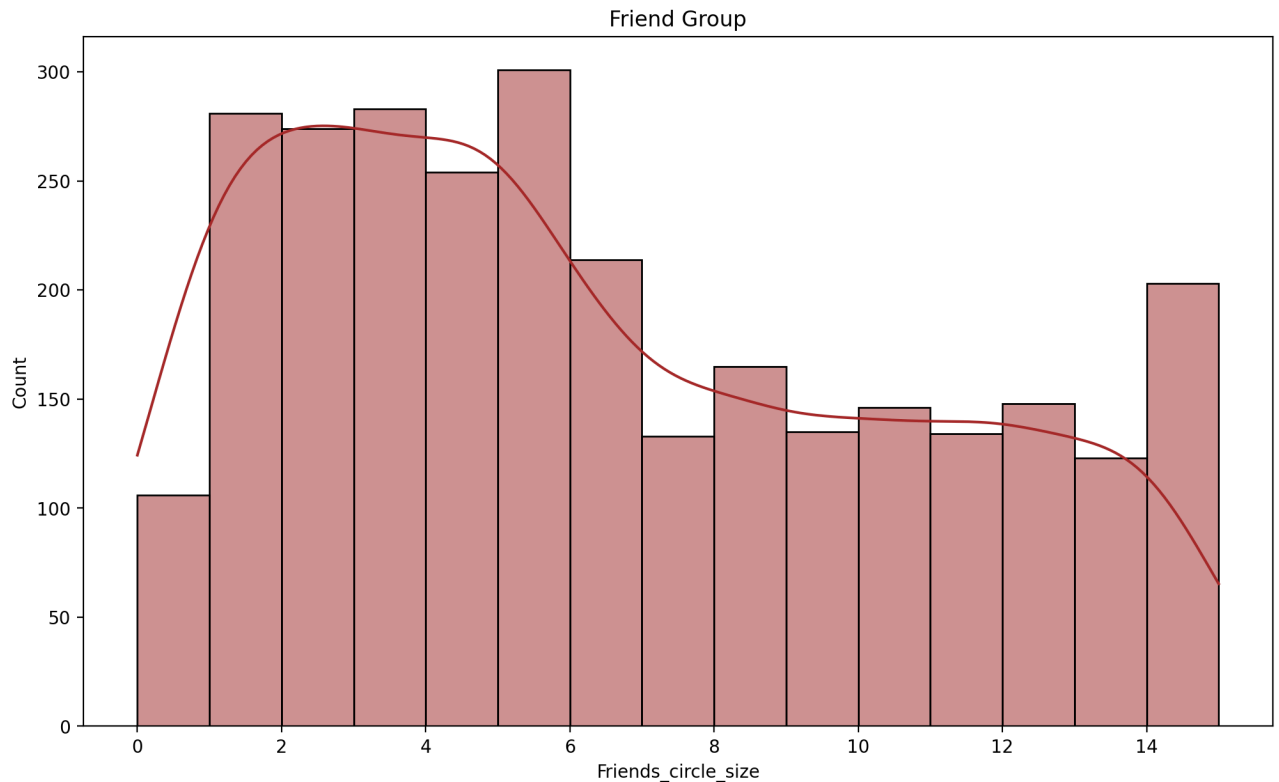
Correlation Heat Map of Numeric Features

The heat map utilizes a color-coded scale to visualize the strength and direction of relationships between various behaviors in the dataset. In this specific visualization, darker shades indicate a strong positive relationship, where two factors increase together, while lighter colors represent values near zero, suggesting little to no connection. A key trend revealed by this map is the very strong negative correlation of -0.718 between time spent alone and social event attendance, meaning that as an individual's solitary time increases, their participation in social gatherings significantly decreases. This is a critical finding because it demonstrates that social behaviors are

not random occurrences; rather, they are deeply and predictably tied to the specific amount of solitary time a person requires to function.

```
sns.histplot(hi_new['Friends_circle_size'],bins=15,color='brown',kde=True)
plt.title('Friend Group')
plt.savefig('Project2_Graphs/hist_friend_size.jpg')
plt.show()
```

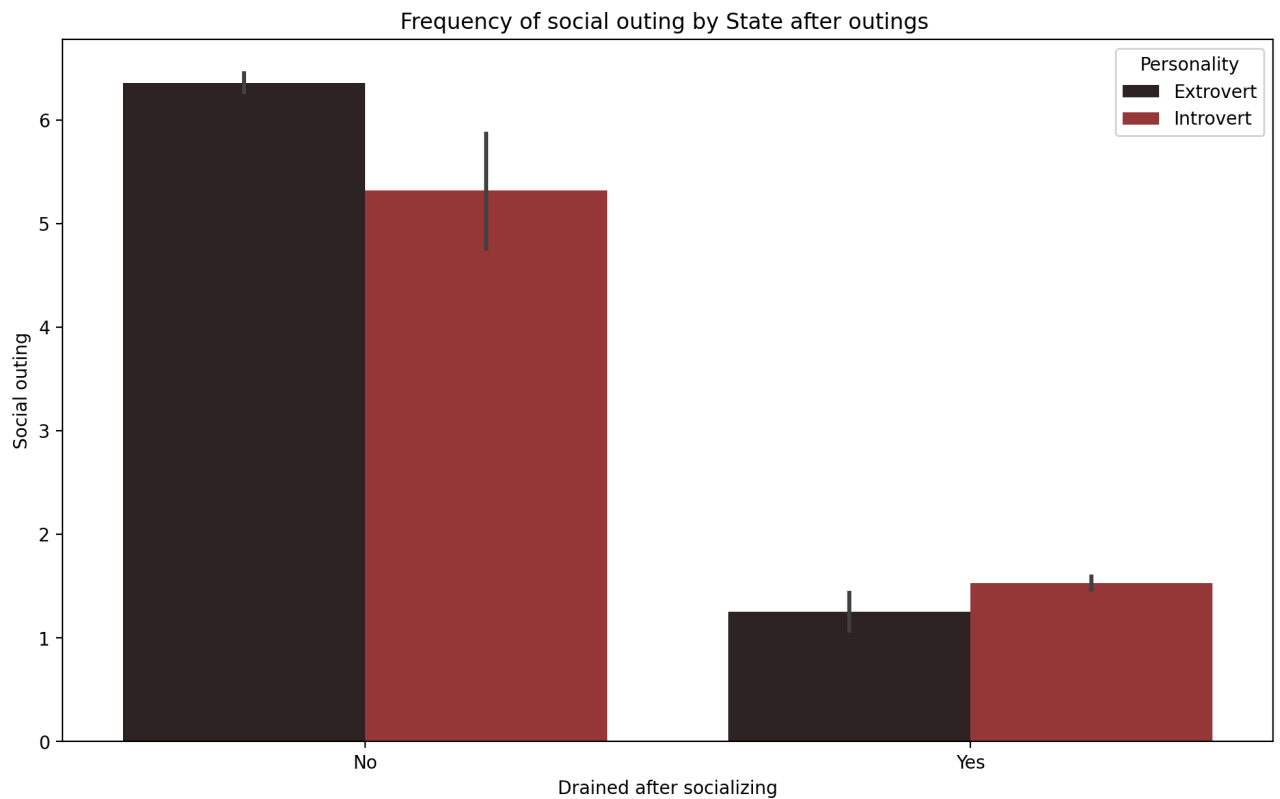
Out [31]:



This Histogram provides a clear look at the distribution and spread of social circles across the 2,900 individuals in this study. The data show that the majority of people have smaller, more intimate friend groups, typically ranging from 1 to 6 friends, which creates a tall cluster on the left side of the graph. However, the visualization also reveals a small but distinct group of "social butterflies" on the far right who maintain much larger circles of 14 to 15 friends. This helps us see that while a smaller inner circle is the standard for most, there is a wide variety in how people define their social networks.

```
sns.barplot(data=hi_new, x='Drained_after_socializing',
y='Social_event_attendance', hue='Personality',palette='dark:brown' )
plt.title('Frequency of social outing by State after outings ')
plt.xlabel('Drained after socializing')
plt.ylabel('Social outing')
plt.savefig('Project2_Graphs/Frequency_of_social_outing
by_State_after_outings.jpg')
plt.show()
```

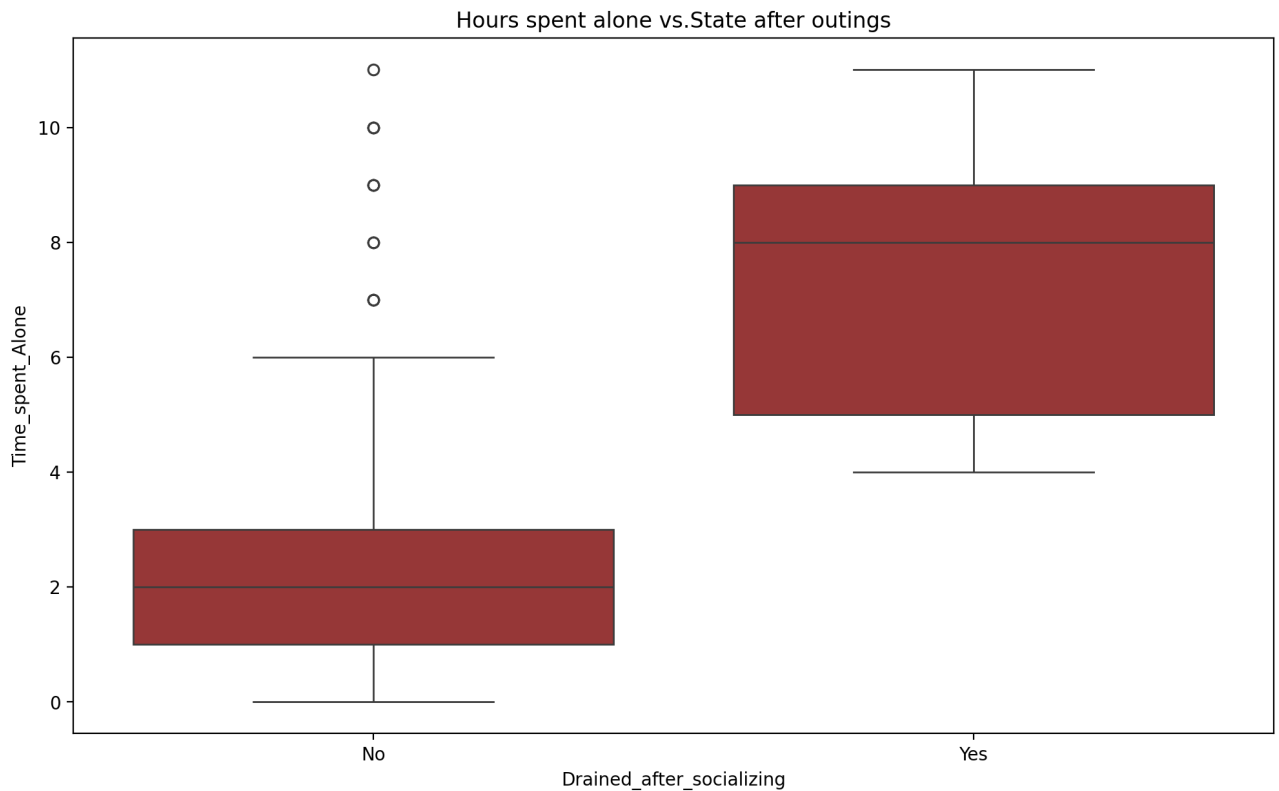
Out [32]:



This Bar Plot highlights the differences between self-identified Introverts and Extroverts. The most striking observation is that Extroverts (represented by the darker bars) consistently attend more social events than Introverts across the board. Furthermore, the chart reveals a significant drop-off in activity for anyone, regardless of personality, who feels drained after socializing. This visualization effectively shows that while personality sets a "baseline" for activity, the internal feeling of exhaustion is a powerful secondary factor that heavily limits how often someone chooses to engage with others.

```
sns.boxplot(hi_new, x='Drained_after_socializing', y='Time_spent_Alone',  
color='brown')  
plt.title('Hours spent alone vs.State after outings')  
plt.savefig('Project2_Graphs/Alone_time_vs._State_after_outings.jpg')  
plt.show()
```

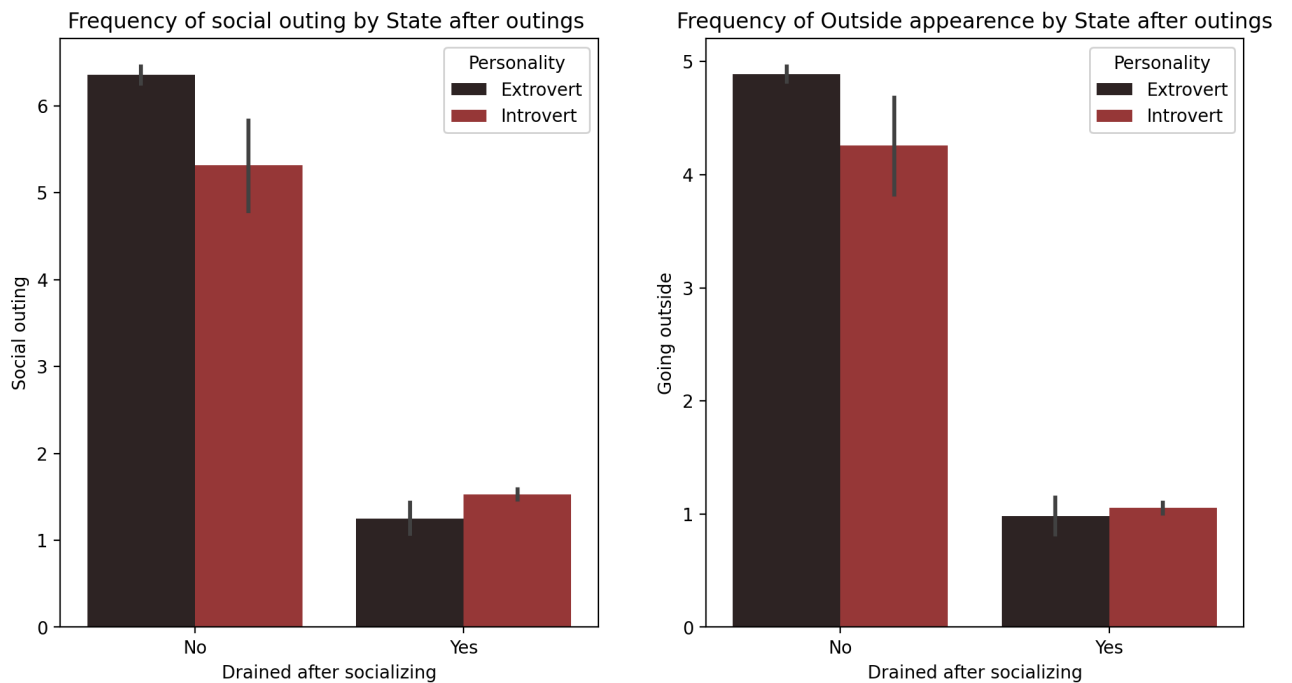
Out [33]:



The Box Plot illustrates the significant difference in solitary time required by different groups based on their energy levels. For those who do not feel drained after socializing, the box is positioned very low, showing they typically spend only one to three hours alone. In contrast, those who do feel drained have a much higher spread, with most spending around eight hours alone to recharge. This visualization also identifies several outliers, individual dots high above the box, representing unique cases where people do not feel drained but still choose to spend an unusually high amount of time alone.

```
fig, axes=plt.subplots(1,2, figsize=(12,6))
sns.barplot(data=hi_new, x='Drained_after_socializing',
y='Social_event_attendance',
hue='Personality',palette='dark:brown',ax=axes[0])
axes[0].set_title('Frequency of social outing by State after outings ')
axes[0].set_xlabel('Drained after socializing')
axes[0].set_ylabel('Social outing')
sns.barplot(data=hi_new, x='Drained_after_socializing',
y='Going_outside', hue='Personality',palette='dark:brown',ax=axes[1])
axes[1].set_title('Frequency of Outside appearance by State after
outings')
axes[1].set_xlabel('Drained after socializing')
axes[1].set_ylabel('Going outside')
plt.savefig('Project2_Graphs/subplots.jpg')
plt.show()
```

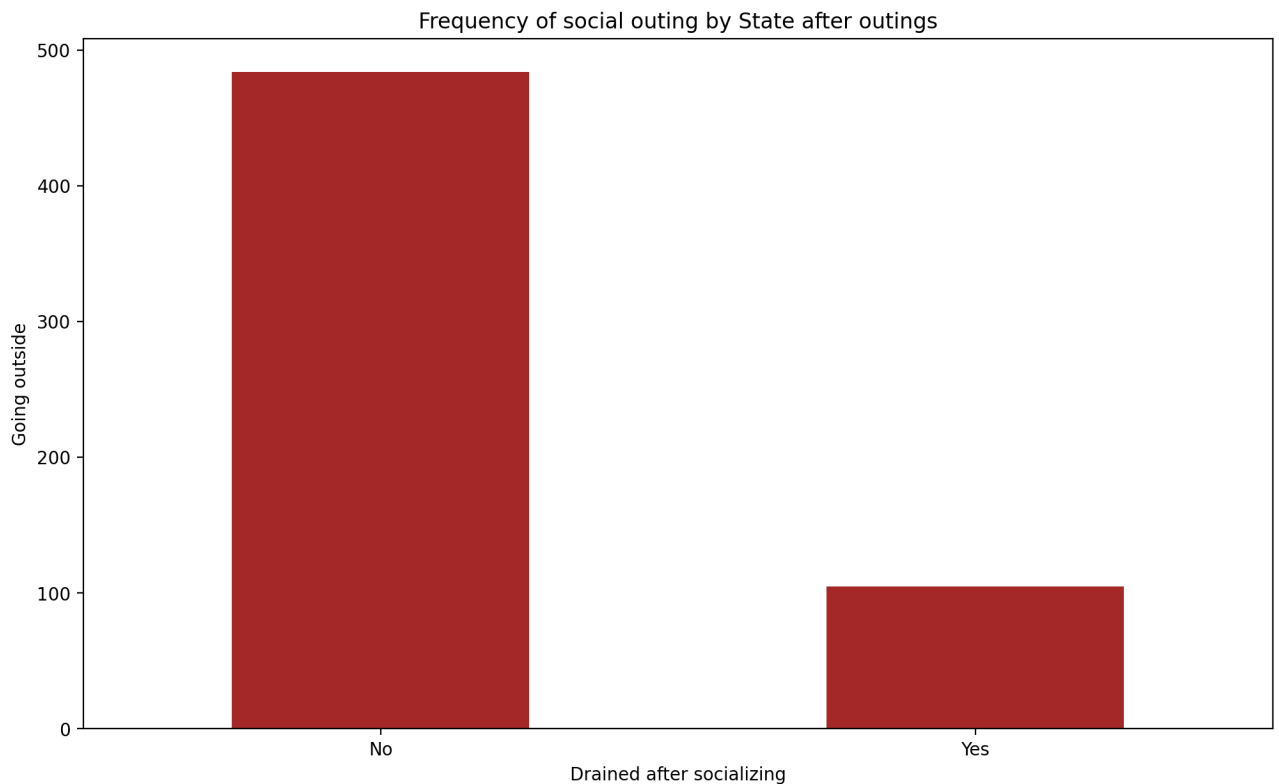
Out [34]:



These side-by-side bar charts provide a comparison that reveals how feeling drained affects different types of movement beyond just parties. The first plot shows that both Introverts and Extroverts significantly reduce their social outings when they feel exhausted. The second plot, 'Frequency of Outside Appearance,' reveals a broader trend: feeling drained doesn't just impact social events; it reduces general activity, such as running errands or just being outdoors. Across both groups, those who are not drained go outside roughly four to five times more often than those who are, suggesting that "social drainage" acts as a major barrier to nearly all forms of movement.

```
state_after_outings = hi_new.groupby('Drained_after_socializing',
observed=True)['Going_outside'].mean() * 100
state_after_outings.plot(kind='bar', color='brown')
plt.title('Frequency of social outing by State after outings')
plt.xlabel('Drained after socializing')
plt.ylabel('Going outside')
plt.xticks(rotation=0)
plt.savefig('Project2_Graphs/group_drained_after_socializing
by_the_by_the_going_outside_avg.jpg')
plt.show()
```

Out [35]:

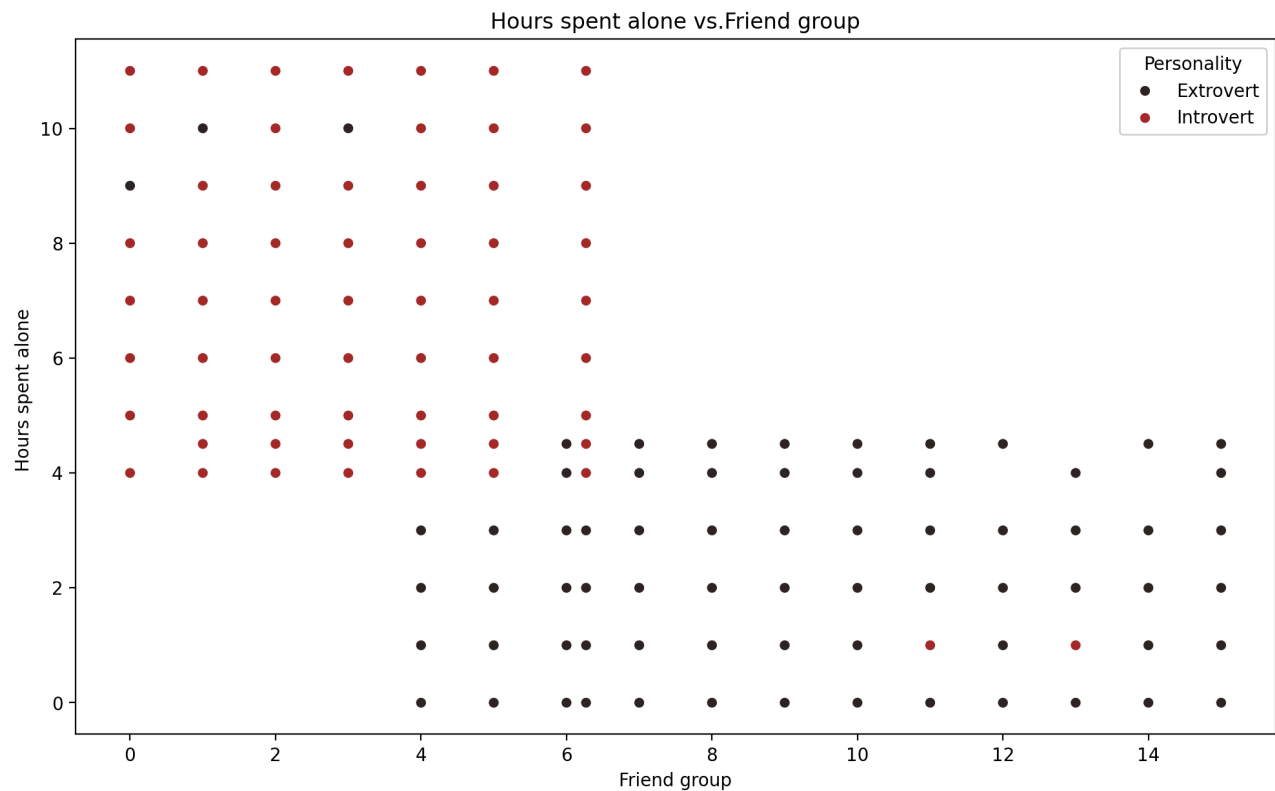


This Bar Chart summarizes the overall frequency of going outside based specifically on a person's state after a social outing. By looking at the average values, it becomes clear that there is a stark divide; the group that feels drained has a much lower overall spread of activity, reaching only a fraction of the height of the non-drained group. This visualization reveals that the state of being drained is perhaps the most influential factor in the entire dataset, as it consistently predicts lower social circles, more time spent alone, and a drastic reduction in how often a person interacts with the world outside their home.

In [36]:

```
sns.scatterplot(data=hi_new, y='Time_spent_Alone',  
x='Friends_circle_size', hue='Personality', palette='dark:brown')  
plt.title('Hours spent alone vs.Friend group')  
plt.ylabel('Hours spent alone ')  
plt.xlabel('Friend group')  
plt.savefig('Project2_Graphs/Hours_spent_alone_vs._Friend_group_scatterplot.jpg')  
plt.show()
```

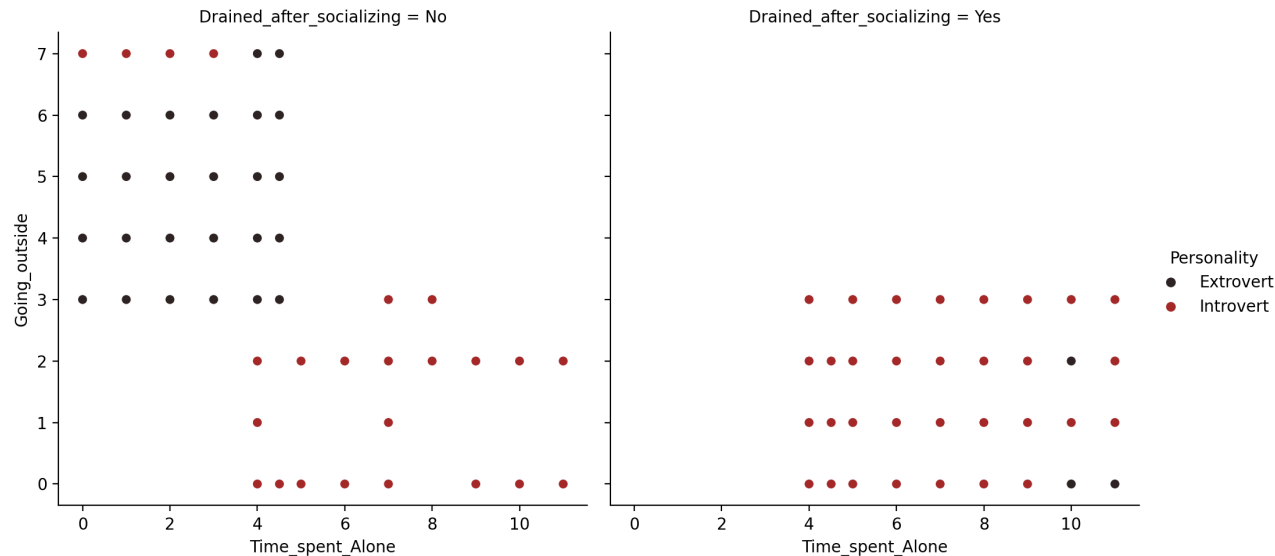
Out [36]:



The Scatter Plot illustrates the relationship between solitude and social circle size, revealing two clear clusters of data points. On one side, there is a dense group representing people who spend upwards of seven hours alone and have very few friends, while the opposite side shows people who spend almost no time alone but have very large social circles. This visualization is important because it reveals a clear trade-off pattern: as the number of friends increases, the hours spent in solitude predictably decrease, showing how these two variables are inversely balanced in daily life.

```
sns.relplot(data=hi_new, x='Time_spent_Alone', y='Going_outside',
hue='Personality', col='Drained_after_socializing', col_wrap=2,
palette='dark:brown')
plt.savefig('Project2_Graphs/Relplot.jpg')
plt.show()
```

Out [37]:



The Relational Plot provides an overview of how solitary time and general activity levels intersect with a person's internal energy state and personality. The downward slope observed across both panels indicates a clear negative trend, revealing that as an individual's time spent alone increases, their frequency of going outside consistently decreases. By splitting the data into two panels based on whether someone feels drained after socializing, the graph highlights a significant shift in behavior: those who feel exhausted are clustered at the top-left, showing high solitude and low activity, while those who do not feel drained maintain a much higher spread of outdoor movement with minimal time alone. Furthermore, the color-coded points show that while Extroverts tend to be more active overall, the state of being drained acts as a major equalizer that forces both personality types toward similar patterns of withdrawal and recovery.

Summary and Conclusion

The data reveals a clear and consistent divide in social behavior driven primarily by personality types and energy management. We observed that the majority of the 2,900 individuals surveyed maintain small, intimate friend groups of one to six people, while a small subset of highly social individuals maintains much larger circles. A central discovery in this analysis is the negative relationship between solitude (alone time) and social engagement. As the time spent alone increases, the frequency of social outings and the size of friend groups decrease predictably. Furthermore, the data highlights that social drainage is a universal factor; regardless of whether someone identifies as an Introvert or an Extrovert, feeling drained after interaction leads to a significant reduction in both social attendance and general outdoor activity.

In conclusion, this study demonstrates that human interaction is not a random occurrence but a balanced system of energy and preference. The data shows that behavior is shaped by a continuous feedback loop between social environments and biological factors. While an individual's personality provides a biological baseline for how they process social stimulation, their state after outings acts as a gatekeeper for future movement. Those who feel biologically recharged by others are far more likely to engage with the world, while those who feel drained require substantial solitary time, often up to eight hours, to achieve homeostasis and recover. Ultimately, these visualizations prove that while environments provide the setting for interaction, our need for energy management is what truly dictates the rhythm of our social lives.

